

Topic 1.5 - Polynomial Functions and Complex Zeros

Guided Notes

Strengthened ECHS edition - reasoning, challenge, and AP-style practice

Name: _____

Class: _____

Date: _____

Why this topic matters

A polynomial's real graph shows only real zeros, but its algebraic structure includes complex zeros as well. Counting multiplicity and using conjugate pairs makes it possible to construct, factor, and audit polynomials while explaining why a degree can exceed the number of visible intercepts.

Learning targets

- Apply the Fundamental Theorem of Algebra and conjugate-pair theorem to count zeros.
- Construct real-coefficient polynomials from real and complex zeros.
- Connect real-zero multiplicity to graph behavior while distinguishing complex zeros from intercepts.

Language and structure

Term	Meaning in this topic
Fundamental Theorem of Algebra	A degree- n polynomial has exactly n complex zeros counted with multiplicity.
conjugate pair	For real coefficients, a nonreal zero $a + bi$ requires $a - bi$.
multiplicity	The number of times a zero occurs as a factor.
real zero	A real input where the polynomial output is zero; an x -intercept.
irreducible quadratic	A real quadratic with negative discriminant and two nonreal conjugate zeros.

Launch - reason before calculating

A degree-5 polynomial with real coefficients has zeros 2, $-1 + i$, and $3i$. List every zero forced by the information and explain why no additional zeros are possible.

Concept 1: Counting zeros and conjugate structure

Core idea

Degree counts all real and nonreal zeros with multiplicity. If coefficients are real, nonreal zeros occur in conjugate pairs because conjugating the equation preserves the real coefficients. Consequently, an odd-degree real polynomial must have at least one real zero.

Worked example - annotate the reasoning

Problem. A degree-6 real polynomial has zeros 4, 4, and $1 - 2i$. What additional zeros are required, and how many zeros remain unspecified?

Model reasoning. The conjugate $1 + 2i$ is required. Counting multiplicity gives four known zeros, so two more zeros remain; they must be real or another conjugate pair.

Check your understanding

Independent

No calculator

A degree-7 real polynomial has zeros -3 (multiplicity 2), $2 + i$, and $5i$. How many zeros remain after including conjugates?

Concept 2: Constructing from complex zeros

Core idea

A conjugate pair $a \pm bi$ creates the real factor $(x - a)^2 + b^2$. Multiply these factors with real linear factors, then use a given point or leading coefficient to determine scale. Factored form often communicates zeros more clearly than expanded form.

Worked example - annotate the reasoning

Problem. Find the monic cubic with zeros 3 and $-2 \pm i$.

Model reasoning. The conjugate pair gives $(x + 2)^2 + 1 = x^2 + 4x + 5$. Thus $P = (x - 3)(x^2 + 4x + 5) = x^3 + x^2 - 7x - 15$.

Check your understanding

Independent

No calculator

Find the monic cubic with zeros 4 and $-1 \pm 2i$.

Concept 3: Multiplicity and the real graph

Core idea

Only real zeros create x -intercepts. At an odd-multiplicity real zero, the sign changes and the graph crosses; at even multiplicity, the sign does not change and the graph touches. Complex factors affect the polynomial's values but do not create real intercepts.

Worked example - annotate the reasoning

Problem. Analyze $P(x) = (x + 1)^2(x - 3)^3(x^2 + 4)$.

Model reasoning. Degree 7. Real zeros: -1 multiplicity 2 (touch) and 3 multiplicity 3 (cross with flattening). Nonreal zeros: $\pm 2i$. Positive leading coefficient and odd degree give left end down, right end up.

Check your understanding

Independent

No calculator

Analyze $(x - 2)^4(x + 5)(x^2 - 2x + 5)$.

Representation laboratory

Connect at least two representations

A real polynomial graph crosses at $x = -4$, touches at $x = 1$, and has no other real intercepts. Its degree is 7. Give two different possible factorizations consistent with the graph and explain how the unseen degrees are accounted for.

Challenge laboratory

Challenge - justify every claim

Construct a real-coefficient quartic whose zeros are $1 \pm i$ and $2 \pm 3i$, and whose value at 0 is 52. Give factored and expanded forms.

Error clinic and calculator strategy

Common errors to diagnose

- Counting only visible x -intercepts as all zeros.
- Forgetting the conjugate of a nonreal zero for real coefficients.
- Ignoring multiplicity when comparing zero count with degree.
- Saying the graph crosses the real axis at a complex zero.

Technology decision

A calculator can approximate complex zeros, but rounded roots should be checked by substitution or reconstructed factors. Numerical output can split repeated roots or hide exact forms, so use algebraic structure whenever coefficients factor cleanly.

AP-style synthesis**Multi-part reasoning task**

A monic degree-5 polynomial with real coefficients has zeros -2 , $1 + 2i$, and 3 with multiplicity 2.

A. List all five zeros counted with multiplicity.

B. Write the polynomial in factored form over the reals.

C. Expand the polynomial and state the only real x -intercepts with cross/touch behavior.

Exit ticket**Before you leave**

1. Why do nonreal zeros come in pairs for real coefficients?

2. What factor corresponds to $4 \pm 3i$?

3. How does even multiplicity affect a graph at a real zero?
